

Individual Poster Presentation (IPP)

Purpose of the IPP

The Individual Poster Presentation (IPP) is an assessed opportunity for each student to demonstrate their personal understanding of their chosen technical discipline within the group design project. Through an A0 printed poster and a short oral presentation, you will:

- Explain the discipline-specific methodology you developed during the project.
- Show how your approach evolved across two levels of model fidelity (e.g., empirical during sprint 2 and physics-based in sprint 3)
- Demonstrate that you understand how your discipline influences the aircraft-level design.
- Present an interdisciplinary sensitivity study to illustrate how your work interacts with another discipline of your choice.
- Defend your work clearly during a 10-minute presentation and Q&A

This assessment tests your individual engineering insight, clarity of technical communication, and ability to contextualise your discipline within the whole-aircraft design process. However, your poster should **not** attempt to summarise the entire aircraft design; instead, it should focus on your chosen discipline and how it interacts with the wider system.

Format

Poster Size: A0 (portrait) – a template format has been provided on Blackboard. Please use a minimum font size of 28pt to ensure readability (including inside figures).

Submission and presentation: Final posters must be submitted to BlackBoard on time to ensure they can be printed for the “poster presentation session”. If your submission is late, you are responsible for printing and delivering your poster on the day of assessment.

During the poster presentation session, you will be given a 1-hour slot during which you must be present at your poster. During the 1-hour slot, you will be visited by at least two assessment panels. Each assessment will consist of:

- a 5-minute presentation delivered in front of your printed poster
- a 5-minute Q&A where the assessors will engage you in technical discussion

What Your Poster Should Contain

The aim of the poster is to clearly communicate your individual understanding of your chosen technical discipline, showing how your methodology developed across two levels of fidelity and how your discipline contributes to, and interacts with, the aircraft-level design. Your poster should prioritise clarity, strong visual communication, and meaningful engineering insight.

Below is a generic description of the suggested content of a poster.

- **Brief Introduction**
 - High-level overview of the design problem to provide context for the reader. E.g., the design mission, your team's chosen concept, and the key challenges for your discipline in implementing this design.
- **Methodology**
 - Describe the methodology used at two stages of the project. 1) at the end of the sprint 2 (e.g., lower-fidelity statistical/empirical methods used to quickly explore the design space). 2) your final methodology
 - Highlight the purpose, limitations, and strengths of each method. Clearly explain how your methodology changed between sprints and why
- **Intradisciplinary Sensitivity Studies**
 - Demonstrate your understanding of how switching from low- to higher-fidelity methods influences both discipline-level predictions and system-level outcomes.
 - Describe the sensitivity of the system to key parameters you have assumed constant (not hyperparameters, e.g., spanwise engine position, spar location, material properties, etc.)
 - Comment on how the assumptions of each method constrain the feasible design space.
- **Interdisciplinary Sensitivity Studies**
 - Include a focused, student-selected analysis demonstrating how your discipline interacts with another discipline of your choice.
 - Include at least one quantitative figure or plot demonstrating the interaction.
 - Explain the physical or analytical mechanism linking the two disciplines and discuss how this relationship influences the broader aircraft design.

For many of the individual disciplines, it will be easy to map this generic format to your own work. In some disciplines, the link is less clear. Please see the notes below for discipline-specific guidance.

Optimisation Engineer

Your suggested focus is on single or multi-objective optimisation. For example, you could explore a DOC-optimised design and an ATR-optimised design in the “low-fidelity section,” then describe a strategy for selecting a multi-objective optimum for the high-fidelity model. For the interdisciplinary sensitivity studies, you can then explore the key sensitivities in your chosen design.

Performance, Propulsion, and Climate

It is suggested that you focus on the climate impact model. For example, your lower-fidelity model could focus solely on CO₂ emissions, whereas your higher-fidelity model could incorporate the impacts of other pollutants, such as sulphates, aircraft-induced cloudiness, and NO_x. The intra- and interdisciplinary sensitivity studies can then follow the suggested format.

Project Manager

The project management poster format will vary slightly to balance your contributions to both the economic discipline model and the broader responsibilities of project management. It is suggested your poster has the following content:

- **Brief Introduction** – as before
- **Methodology** – explanation of a single-fidelity, i.e., your final direct operating cost (DOC) model.

- **Intra- and interdisciplinary study** – as before, minus comparison of fidelities.
- **Project Management Strategy and Reflection**
 - provide an overview of your chosen project management methodology
 - give semi-quantitative examples of how project health/progression was monitored
 - Reflect on how processes were changed between sprints to improve monitoring and progression

What the Panel is Looking For

A strong poster will demonstrate that you:

- Have a deep understanding of your discipline tool, its assumptions, and its limitations.
- Understand the progression from simple (classical/statistical) to physics-based methods.
- Can clearly explain how your discipline affects the aircraft-level design.
- Have performed a meaningful interdisciplinary analysis.
- Can communicate visually with clarity (e.g., clean plots, limited text, intuitive layout).
- Can confidently answer questions about your modelling choices and sensitivities.

Final Advice

The goal of the IPP is to convince the panel that you have a clear understanding of your chosen technical discipline, understand its relationship with the wider aircraft design, and have built your methodology upon clear assumptions and sound engineering judgment. To achieve this, it is important to:

- Be clear rather than comprehensive – don't present/rush through too much content
- State your assumptions – a successful student will know the limitations of their modelling assumptions
- Use simple diagrams and sketches where possible – help the panel visualise what you are talking about

Overall, if you show good engineering judgement and an honest reflection of the challenges, you are on course to do well in this unit.

Times, dates, and locations of IPP presentations, as well as deadlines, will be provided on the main BB page.